KATLEGO MOKOTO

AI Engineer · AI Systems Architect · Agentic / LLM Developer

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Status: AVAILABLE

SUMMARY

AI engineer and developer — products, platforms, and agentic systems.

Targeting AI engineering / AI developer roles at OpenAI, Anthropic, Google DeepMind, Meta AI, Microsoft AI, NVIDIA, and applied-AI startups (xAI, Waymo, ElevenLabs, Scale AI, Sierra, Cursor, Perplexity, Cognition, Replit, Mercor).

Portfolio proof: Mens Mercatus Sound Deck, Forex Mercatus Codex, JobNet, Flutter overlay, Antigravity / local LLM stack.

TECHNICAL STACK

Languages: Python, TypeScript, JavaScript, SQL, Bash, Solidity

Frontend: React, Next.js, Flutter, Tailwind, Web Audio / DAW UIs

AI / ML: LLM orchestration, RAG, LangChain, Ollama, Agent Zero, prompt/tool agents, ChromaDB, local inferencing (Apple Silicon M4)

Backend: FastAPI, Node.js, Express, Django, WebSockets, Redis

Data: PostgreSQL, Pandas, NumPy, yfinance, OR-Tools

Cloud & Infra: Nebius, Docker, Linux, CI/CD, AWS, edge / local-first AI

Audio / Realtime: Web Audio API, mic capture, stem DSP, generative synthesis

Systems: P2P/BLE mesh, agentic workflows, Cursor IDE automation

CORE DOMAINS

Agentic AI & Local Compute

Design and ship sovereign agent stacks: tool-using agents, RAG over live data, local LLM orchestration to cut API dependency, and high-throughput inferencing on Apple Silicon.

OLLAMA · AGENT ZERO · CURSOR · RAG · CHROMADB · APPLE SILICON M4

Applied AI Product Engineering

Build end-to-end AI products (generation UIs, copilots, job-automation meshes) with production-minded UX, offline-capable clients, and clear human-in-the-loop controls.

FULL-STACK · WEB AUDIO · AUTOMATION · USER-FACING AI TOOLS

Quantitative & Realtime Systems

Fuse market data pipelines with LLMs and interactive desks for analysis, alerting, and decision support under latency and reliability constraints.

LANGCHAIN · YFINANCE · HIGHCHARTS · PYTHON DESKS

SELECTED AI PORTFOLIO

SoundX / Sound Deck — 2026 · Browser DAW / AI music [DEMO]

Original browser DAW with AI Producer (order-taking studio assistant), demo stem split, 16-step beats, mic/vocal pipelines, VST slot framework, and in-house text-to-music / stem / continuation generation.

- Generate, demo play, mic, and AI Producer on mensmercatus.online apps hub.
- Gold/black/white Mens Mercatus product UI; no third-party music-site dependency.

Web Audio, Agents, DSP, TypeScript/JS

JobNet — 2026 · Career automation [DEMO]

Weekday pipeline: noon multi-board scrape; CV tailor + AI Hawk draft fills with human confirm before submit; Gmail reply watch + SMS/email notify.

- Noon scrape across top AI employers + early-stage trackers.
- 3 AM CV tailor + draft fills; human-in-the-loop gates before any submit.

Automation, Agents, Gmail, macOS shell

Mercatus Cartel — 2026 · Event-mode pack [DEMO]

Screens, schematics, and compliance gates on the site. Assist LIVE: same-origin /api/cartel/ → VPS robot.

- Event-mode product pack with architecture diagrams and compliance gates.
- Human-in-the-loop assist path with honest DEMO labels.

Event systems, Compliance, Assist, UI systems

Forex Mercatus Codex — 2026 · Python desk :8790 [DEMO]

~87MB Python + Highcharts desk for FX workflows; Chart, Multi, Whales, Flow, Bot. Hosted on VPS as live paper research.

- Inventory and load-path optimization for faster cold start and clearer static serving.
- Market data fused with interactive analysis desks.

Python, Highcharts, yfinance, Paper only

Project Antigravity — Mar 2026 – Present

AI Systems Architect & Founder

- Sovereign, decentralized AI OS for autonomous task execution and specialized agentic workflows.
- Terminal-first AI tooling on M4 with Ollama; Nebius for scalable training/deploy of custom models; Agent Zero + Cursor orchestration.

Ollama, Agent Zero, Nebius, LLM Ops

Mens Mercatus Ecosystem — Ongoing · mensmercatus.online

Full-stack & AI trading systems

- Telegram AI trading assistant: RAG + live financial feeds for XAUUSD / cyclical models (SMT, Gann frameworks) with institutional-order context.
- Brand, UI/UX, and static portfolio architecture for the Mens Mercatus digital ecosystem.

Python, LangChain, ChromaDB, Next.js

Messmakers / Flutter Overlay — 2026

Mobile UI systems

- Flutter package fixing z-order: gray AI sound/volume bar above white social band; theme extension, haptics, demo app — drop-in for Messmakers / ThinkPulse clients.

Flutter, Dart, UI systems

ProxNet · Infrastructure.log — Parallel systems work

Edge & mesh

- BLE/WebRTC mesh social (offline-first) and M4 SSD arrays for model weights / vector DBs — edge-ready AI compute hygiene.

React Native, P2P, Edge

SELECTED EXPERIENCE NARRATIVE

- Hands-on builder of agentic products: copilots that execute studio/job/market actions with explicit user confirmation gates.
- Comfortable in Python and TypeScript full-stack; fast iteration with Cursor, local models, and cloud GPUs when needed.
- Background in fleet/logistics optimization (OR-Tools) and diesel/mechanical operations — strong systems thinking under real constraints.

Portfolio: mensmercatus.online · Extended careers pack available on request · Homelake / Randfontein · Open to remote / global AI eng roles.